

Supervised Learning: Classification, Part I

Armeen Taeb & Ali Shojaie

July 22-24, 2026

Summer Institute in Statistics for Big Data
University of Washington

Classification

- ▶ Regression involves predicting a continuous-valued response.

Classification

- ▶ Regression involves predicting a continuous-valued response.
- ▶ Classification involves predicting a categorical / qualitative response:
 - ▶ Cancer versus Normal
 - ▶ Tumor Type 1 versus Tumor Type 2 versus Tumor Type 3

Classification

- ▶ Regression involves predicting a continuous-valued response.
- ▶ Classification involves predicting a categorical / qualitative response:
 - ▶ Cancer versus Normal
 - ▶ Tumor Type 1 versus Tumor Type 2 versus Tumor Type 3
- ▶ Classification problems tend to occur even more frequently than regression problems in biomedical applications.

Classification

- ▶ Regression involves predicting a continuous-valued response.
- ▶ Classification involves predicting a categorical / qualitative response:
 - ▶ Cancer versus Normal
 - ▶ Tumor Type 1 versus Tumor Type 2 versus Tumor Type 3
- ▶ Classification problems tend to occur even more frequently than regression problems in biomedical applications.
- ▶ Just like regression,
 - ▶ Classification cannot be blindly performed in high-dimensions **because you will get zero training error but awful test error**;
 - ▶ Properly estimating the test error is crucial; and
 - ▶ There are a few tricks to extend classical classification approaches to high-dimensions, which we have already seen in the regression context!

Classification

- ▶ Categorical / qualitative variables take values in an unordered set: e.g.
 $\text{eye color} \in \{\text{brown}, \text{blue}, \text{green}\}$
 $\text{email} \in \{\text{spam}, \text{not spam}\}.$
- ▶ We want to build a function that takes as input the feature vector X and predicts the value for Y .
- ▶ Often we are more interested in estimating the **probability** that X belongs to a given category.
- ▶ For example: we might want to know the probability that someone will develop diabetes, rather than to predict whether or not they will develop diabetes.

Can't We Just Use Linear Regression?

- Classify an emergency room patient on the basis of her symptoms to one of three conditions:

$$Y = \begin{cases} 1 & \text{if stroke;} \\ 2 & \text{if drug overdose;} \\ 3 & \text{if epileptic seizure.} \end{cases}$$

- If we apply linear regression, then the results will depend on the choice of coding . . . and the coding implies an ordering among the medical conditions.
- A classification approach is more appropriate.

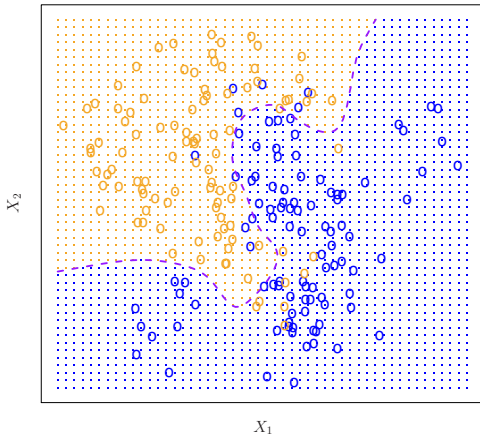
Classification

- ▶ There are many approaches out there for performing classification.
- ▶ We will discuss 3: k-nearest neighbors, logistic regression, and support vector machines.

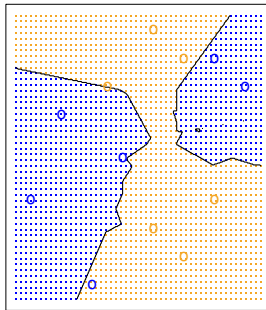
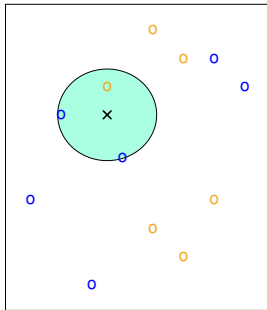
K -Nearest Neighbors

- ▶ Can I take a totally non-parametric (model-free) approach to classification?
- ▶ **K -nearest neighbors:**
 1. Identify the K observations whose X values are closest to the observation at which we want to make a prediction.
 2. Classify the observation of interest to the most frequent class label of those K nearest neighbors.

K-Nearest Neighbors

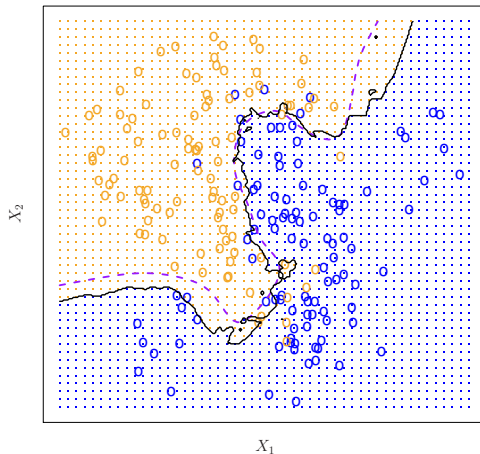


K-Nearest Neighbors



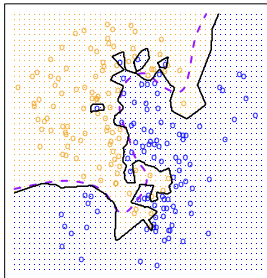
K-Nearest Neighbors

KNN: K=10

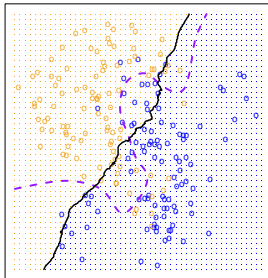


K-Nearest Neighbors

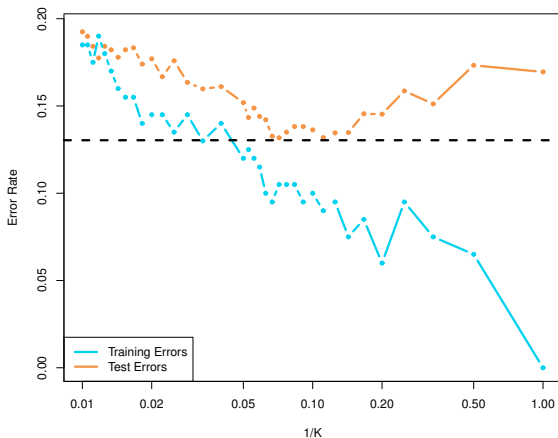
KNN: K=1



KNN: K=100



K-Nearest Neighbors



K-Nearest Neighbors

- ▶ Simple, intuitive, model-free.
- ▶ Good option when p is very small.
- ▶ Curse of dimensionality: when p is large, no neighbors are “near”. All observations are close to the boundary.
- ▶ **Do not use in high dimensions!**

Logistic Regression

- ▶ Logistic regression is the straightforward extension of linear regression to the classification setting.

Logistic Regression

- ▶ Logistic regression is the straightforward extension of linear regression to the classification setting.
- ▶ For simplicity, suppose $y \in \{0, 1\}$: a two-class classification problem.

Logistic Regression

- ▶ Logistic regression is the straightforward extension of linear regression to the classification setting.
- ▶ For simplicity, suppose $y \in \{0, 1\}$: a two-class classification problem.
- ▶ The simple linear model $y = X\beta + \epsilon$ doesn't make sense for classification.

Logistic Regression

- ▶ Let $p(X) = \Pr(Y = 1|X)$.
- ▶ Suppose we want to use **biomarker level** to predict **probability of cancer**.
- ▶ Logistic regression uses the form

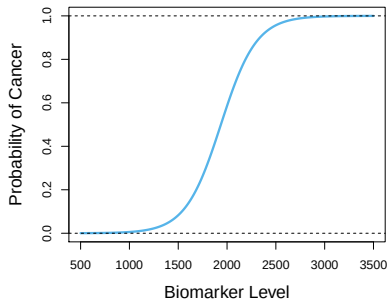
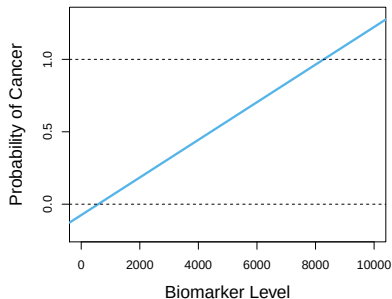
$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}.$$

- ▶ $p(X)$ will lie between 0 and 1.
- ▶ Furthermore,

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X.$$

- ▶ This function of $p(X)$ is called the **logit** or **log odds**.

Why Not Linear Regression?



- ▶ Left: linear regression.
- ▶ Right: logistic regression.

Multiple Logistic Regression

- Just like before:

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}.$$

- And just like before:

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p.$$

Example in R

```
xtr <- matrix(rnorm(1000*20),ncol=20)
beta <- c(rep(1,10),rep(0,10))
ytr <- 1*((xtr%*%beta + .2*rnorm(1000)) >= 0)
mod <- glm(ytr~xtr,family="binomial")
print(summary(mod))
```

Three Ways to Extend Logistic to High Dimensions

Three Ways to Extend Logistic to High Dimensions

1. Variable Pre-Selection
2. Ridge Logistic Regression
3. Lasso Logistic Regression

Three Ways to Extend Logistic to High Dimensions

1. Variable Pre-Selection
2. Ridge Logistic Regression
3. Lasso Logistic Regression

How to decide which approach is best, and which tuning parameter value to use for each approach? **Cross-validation** or **validation set approach**.

What is an appropriate validation measure?

For classification without a probability or score:

- Misclassification rate:

$$\frac{\text{\#test samples misclassified}}{\text{total \# of test samples}}$$

What is an appropriate validation measure?

For probabilistic classification

- ▶ Can still use misclassification rate.
- ▶ Like in continuous regression could use SSE:

$$\sum_{i \in \text{test}} (y_i - \hat{p}_i)^2$$

- ▶ Often preferable to use “predictive [log]likelihood”:

$$-\log \left[\prod_{i \in \text{test}} \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i} \right]$$

- ▶ Can also use ROC-curve-based metric (eg. AUC)

Remember though; all of these must be conducted on a **separate validation set**.

Example in R: Lasso Logistic Regression

```
xtr <- matrix(rnorm(1000*20),ncol=20)
beta <- c(rep(1,5),rep(0,15))
ytr <- 1*((xtr%*%beta + .5*rnorm(1000)) >= 0)
cv.out <- cv.glmnet(xtr, ytr, family="binomial", alpha=1)
plot(cv.out)
```

Let's Try It Out in R!

Chapter 4 R Lab
Skip part on LDA & QDA
www.statlearning.com

Support Vector Machines

- ▶ Developed in around 1995.

Support Vector Machines

- ▶ Developed in around 1995.
- ▶ Touted as “overcoming the curse of dimensionality.”

Support Vector Machines

- ▶ Developed in around 1995.
- ▶ Touted as “overcoming the curse of dimensionality.”
- ▶ Does not automatically overcome the curse of dimensionality!!!

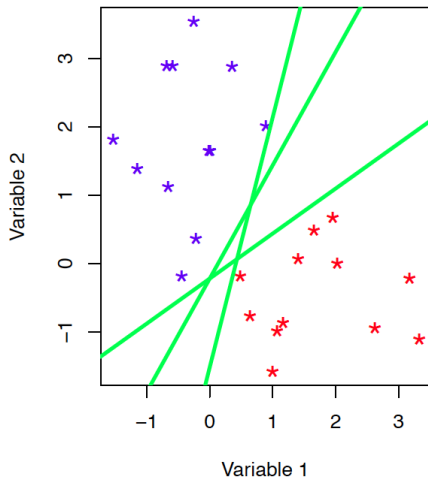
Support Vector Machines

- ▶ Developed in around 1995.
- ▶ Touted as “overcoming the curse of dimensionality.”
- ▶ Does not automatically overcome the curse of dimensionality!!!
- ▶ Fundamentally and numerically very similar to logistic regression.

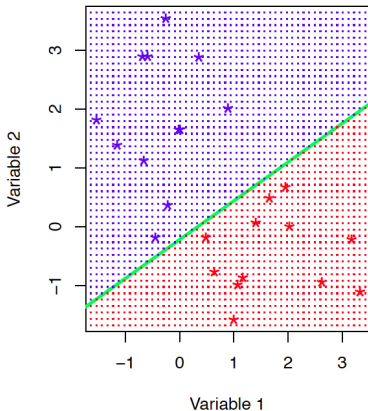
Support Vector Machines

- ▶ Developed in around 1995.
- ▶ Touted as “overcoming the curse of dimensionality.”
- ▶ Does not automatically overcome the curse of dimensionality!!!
- ▶ Fundamentally and numerically very similar to logistic regression.
- ▶ But, it is a nice idea.

Separating Hyperplane

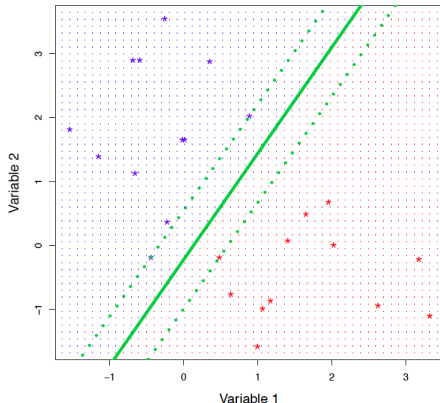


Classification Via a Separating Hyperplane



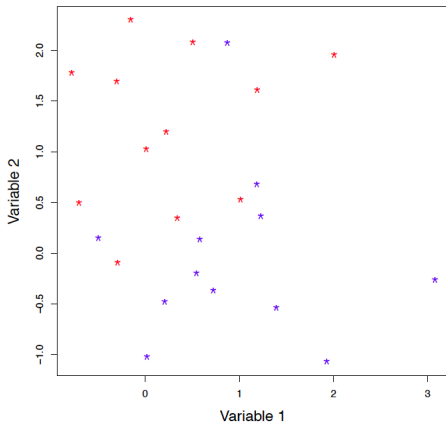
Blue class if $\beta_0 + \beta_1 X_1 + \beta_2 X_2 > c$; red class otherwise.

Maximal Separating Hyperplane

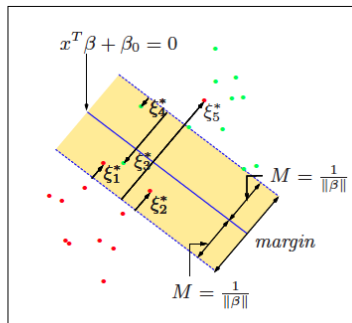
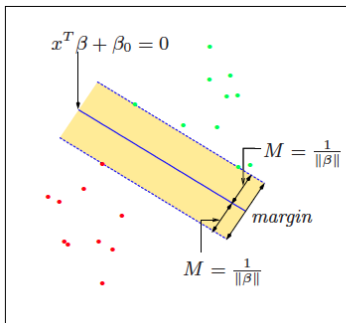


Note that only a few observations are **on the margin**: these are the **support vectors**.

What if There is No Separating Hyperplane?



Support Vector Classifier: Allow for Violations



Support Vector Machine

- The support vector machine is just like the support vector classifier, but it elegantly allows for non-linear expansions of the variables: “non-linear kernels”.

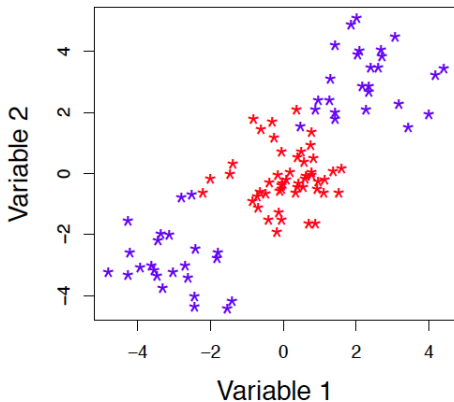
Support Vector Machine

- ▶ The support vector machine is just like the support vector classifier, but it elegantly allows for non-linear expansions of the variables: “non-linear kernels”.
- ▶ However, linear regression, logistic regression, and other classical statistical approaches can also be applied to non-linear functions of the variables.

Support Vector Machine

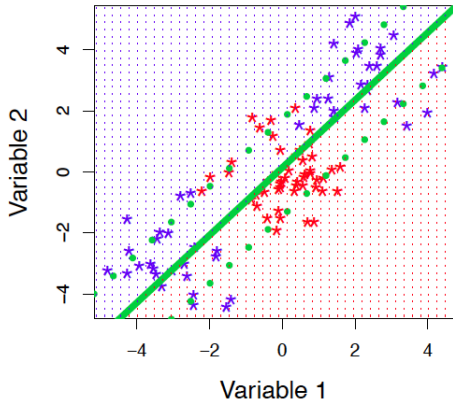
- ▶ The support vector machine is just like the support vector classifier, but it elegantly allows for non-linear expansions of the variables: “non-linear kernels”.
- ▶ However, linear regression, logistic regression, and other classical statistical approaches can also be applied to non-linear functions of the variables.
- ▶ For historical reasons, SVMs are more frequently used with non-linear expansions as compared to other statistical approaches.

Non-Linear Class Structure



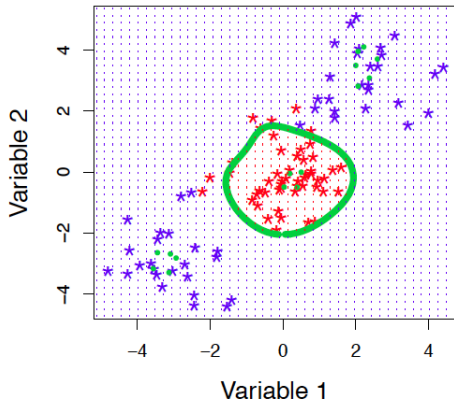
This will be hard for a linear classifier!

Try a Support Vector Classifier



Uh-oh!!

Support Vector Machine



Much Better.

Is A Non-Linear Kernel Better?

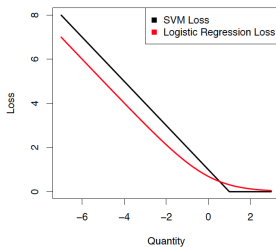
Is A Non-Linear Kernel Better?

- Yes, if the true decision boundary between the classes is non-linear, and you have enough observations (relative to the number of features) to accurately estimate the decision boundary.

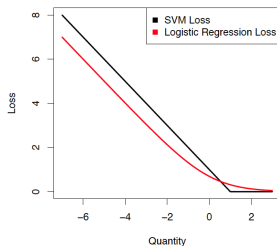
Is A Non-Linear Kernel Better?

- ▶ **Yes**, if the true decision boundary between the classes is non-linear, and you have enough observations (relative to the number of features) to accurately estimate the decision boundary.
- ▶ **No**, if you are in a very high-dimensional setting such that estimating a non-linear decision boundary is hopeless.

Support Vector Classifier Versus Logistic Regression

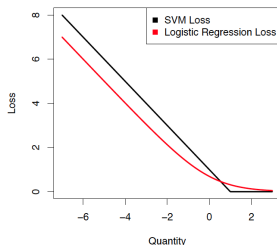


Support Vector Classifier Versus Logistic Regression



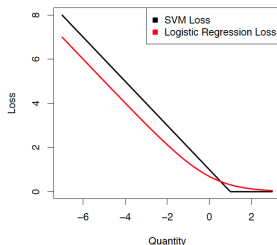
- Bottom Line: Support vector classifier and logistic regression aren't that different!

Support Vector Classifier Versus Logistic Regression



- ▶ Bottom Line: Support vector classifier and logistic regression aren't that different!
- ▶ Neither they nor any other approach can overcome the “curse of dimensionality”.

Support Vector Classifier Versus Logistic Regression



- ▶ Bottom Line: Support vector classifier and logistic regression aren't that different!
- ▶ Neither they nor any other approach can overcome the “curse of dimensionality”.
- ▶ SVM uses a non-linear kernel... but could do that with logistic or linear regression too!

In High Dimensions...

In High Dimensions...

- ▶ In SVMs, a tuning parameter controls the amount of flexibility of the classifier.

In High Dimensions...

- ▶ In SVMs, a tuning parameter controls the amount of flexibility of the classifier.
- ▶ This tuning parameter is like a **ridge penalty**, both mathematically and conceptually. The SVM decision rule involves all of the variables.

In High Dimensions...

- ▶ In SVMs, a tuning parameter controls the amount of flexibility of the classifier.
- ▶ This tuning parameter is like a **ridge penalty**, both mathematically and conceptually. The SVM decision rule involves all of the variables.
- ▶ Can get a **sparse** SVM using a **lasso penalty**; this yields a decision rule involving only a subset of the features.

In High Dimensions...

- ▶ In SVMs, a tuning parameter controls the amount of flexibility of the classifier.
- ▶ This tuning parameter is like a **ridge penalty**, both mathematically and conceptually. The SVM decision rule involves all of the variables.
- ▶ Can get a **sparse** SVM using a **lasso penalty**; this yields a decision rule involving only a subset of the features.
- ▶ Logistic regression and other classical statistical approaches could be used with non-linear expansions of features. But this makes high-dimensionality issues worse.

Let's Try It Out in R!

Chapter 9 R Lab
www.statlearning.com

Discussion Questions

Suppose someone came to a statistical consulting service you were running and said...

I want to try and classify patients as having breast cancer, or not based on gene expression in serum.

I'm pretty excited because I just found two awesome datasets:

The first, from the Farnsworth Lab, has serum expression measured using RNA-seq in 5000 patients with breast cancer;

The second, from the Wernstrom Lab, has serum expression measured using microarrays on 5000 healthy patients.

I wanted to combine them to build my classifier

What concerns, if any, come to mind?

Discussion Questions

Suppose we want to classify patients as having cancer/not having cancer using methylation on cf-dna fragments

In particular, say we initially consider 10000 cpg sites, and try to build a classification model that uses proportion of methylated fragments at each of those sites as features.

Would it make sense to run an SVM with a non-linear kernel here?

If we used cross-validation to select between both that SVM, and a LASSO-logistic regression, what might happen?